

닌자의 신 — Master Game Design Document

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사람용 다운로드: Master Production GDD PDF — 이 Markdown 정본에서 생성한 읽기 전용 snapshot
발행/검수 계약: <code>docs/PDF_EXPORT.md</code> 와 <code>docs/publication/NINJA_SURVIVAL_MASTER_GDD_PDF_MANIFEST.json</code> ; PDF 존재는 runtime-Human Usability-Player Experience 검증을 뜻하지 않는다.
정본 우선순위: 최신 사용자 승인 → <code>AGENTS.md</code> → <code>CURRENT_CONFIRMED_DECISIONS.md</code> → dated canon → <code>ACTIVE_CONTEXT.md</code> → actual code/data/Scene/test → Base → benchmark
범위: 이 파일은 사람이 읽는 GDD와 AI가 검색하는 구현 계약의 공통 정본이다. DEC-035는 기존 Notion 구조와 현재 작업물을 repository migration archive에 보존한 뒤, repository-only owner로 전환한다.

00. Canon Snapshot

■■■■ ■은 Godot 4.x/GDScript로 만드는 2D top-down survival roguelike다. 플레이어는 하나의 고정 닌자 신분으로 네 유파의 전장을 한 번씩 고르고, 각 유파가 가르치는 서로 다른 위험 처리법을 몸으로 익힌 뒤, 공간 배치형 백팩과 Fate를 조합해 최종 재앙에 맞설 자신만의 인법을 만든다.

상태 표기

상태	뜻	이 문서에서의 사용
DOCUMENTED	문서에 정의됨	미래 product contract
CONFIRMED	사용자 승인 정보	구현이 따라야 하는 rule
IMPLEMENTED	<code>main</code> code/Scene/data에 존재	source path로 확인
AUTOMATED_TEST_PASS	해당 범위 GUT/CI evidence 존재	Human 품질을 뜻하지 않음
RUNTIME_VERIFIED	실행 render/input evidence가 있음	범위와 SHA를 별도로 표시
UX_VERIFIED	실제 Human usability evidence 있음	현재 대부분 NOT_RUN

상태	뜻	이 문서에서의 사용
RELEASE_READY	platform/product quality gate 통과	현재 NOT_READY

지금의 Work 5 위치

Phase 1 ■■/■■■: IN PROGRESS
Phase 2 preproduction / Definition of Ready: NOT STARTED
Phase 3 asset production: PARTIAL (■■ ■■■ asset■)
Phase 4 Godot implementation: T12~T16 machine baseline merged; current package NOT STARTED
Phase 5 Human vertical-slice validation: NOT_RUN

01. Source Registry

ID	source	역할	사용 상태
SRC-REP0-01	AGENTS.md	engine, authority, protected boundaries	CONFIRMED
SRC-REP0-02	docs/CURRENT_CONFIRMED_DECISIONS.md	current product/visual/evidence ledger	CONFIRMED
SRC-REP0-03	docs/ACTIVE_CONTEXT.md	mutable resume route	CONFIRMED
SRC-REP0-04	DEC-014~025, DEC-026~034	product, encounter, current decisions	CONFIRMED
SRC-REP0-05	scripts/, scenes/, assets/, tests/	actual implementation reality	IMPLEMENTED where paths exist
SRC-REP0-06	docs/visual/*, asset manifests	visual grammar and consumer evidence	mixed; see AST- *
SRC-NOTION-01	former Human Home / Core Systems / Visual Bible / Production Handoff	preserved migration snapshot and attachment provenance	docs/migration/notion/; not active canon
SRC-EXT-01	official/Steam pages, researched 2026-08-28	positioning references	REFERENCE_ONLY

Current conflict register

conflict ID	current canonical claim	actual source reality	disposition
CON-01	Trace is an in-world recovery/gate object	MainController currently exposes R/button recovery	IMPLEMENTATION_REQUIRED — canonical Trace behavior wins
CON-02	icon-first statuses; only struck enemy HP bar	EnemyEffectBadge is text and no enemy HP presentation exists	IMPLEMENTATION_REQUIRED
CON-03	school-owned Core/Elite/Boss compositions	current scene instantiates generic enemy/Boss representations	IMPLEMENTATION_REQUIRED

conflict ID	current canonical claim	actual source reality	disposition
CON-04	Workbench is a full spatial decision	current RestFlowUI exposes route/Fate but not Boss reward selection or 6×6 placement	BLOCKING_GAP
CON-05	DEC-033 economy / persistent Soul settlement	current RunBuildState grants normal 1G and Boss 25G; no Elite, rank, wallet, or settlement owner	IMPLEMENTATION_REQUIRED
CON-06	true Run-end Soul settlement	first Human endpoint stops at fourth-Boss final_binding_eligible before Final Binding	CONFIRMED — no Soul settlement at that endpoint; true conclusion package owns success settlement

02. One-page Vision

Player Promise — PX-001

“내가 선택한 유파의 위험 처리법으로 전장을 읽고, 네 전승을 하나의 닌자식으로 조립해 재앙을 봉인한다.”

read threat / position
-> automatic school action reveals a school-specific opportunity
-> choose route, reward, spatial placement, Fate
-> commit one build and one next danger together
-> survive a harder school with the consequences
-> remember the personal ninja method that emerged

Design Pillars

ID	pillar	player value	guardrail
PIL-01	위험을 다르게 다루는 네 유파	같은 적도 다른 위치·우선순위·준비로 풀린다	유파를 원소 스킨으로 축소하지 않는다
PIL-02	자동전투 안의 읽는 이동	조준 부담 없이 위치와 군집으로 실력 표현	자동전투를 완전 방치형으로 만들지 않는다
PIL-03	공간 백팩이 만든 인법	무엇을 얻고 어디에 놓을지의 고민이 전투력으로 읽힌다	UI가 legality/economy/combat authority를 소유하지 않는다
PIL-04	네 전승을 쌓는 한 명의 닌자	route clear order와 visual layers가 개인적 전설을 만든다	네 주인공/네 독립 게임으로 분열하지 않는다
PIL-05	빠르고 정직한 피드백	위험→선택→반응→보상/실패 학습이 읽힌다	text badge, 숨은 rule, 장식이 전투 가독성을 이기지 않는다

차별점 / 판매 포인트

- USP-01 유파 선택이 빌드 색상이 아니라 위험 처리 방식이다. 봉마는 이동형 진지, 천술은 공간 반응, 귀인은 위험한 근접 지속, 흑영은 위험 우선 처형을 가르친다.
- USP-02 자동 사격 survival과 spatial Workbench를 한 Run의 원인-결과로 묶는다. Shop/Chest/Boss reward는 즉시 파워가 아니라 배치·인접·조합·Fate commit을 거쳐 전투 권한이 된다.
- USP-03 하나의 닌자가 네 전승을 누적한다. 선택한 시작 유파만 강한 Stage-3 시각 층이 되고, 나머지는 읽히는 보조 층으로 결합한다.

03. Player Experience: 첫 5·15·30분

시간	player experience	성공 신호	실패가 남길 학습	current status
0-30초	선택 유파의 위험 처리 언어를 본다	학교 feedback, 움직임, 자동행동 결과	“무엇을 보고 어디로 움직여야 하는가”	PARTIAL; optional help exists but unassisted UX NOT_RUN
0-5분	Core 압박→Elite→Trace→Boss의 긴장을 경험	Elite token/Trace, Boss warning, victory result	Trace를 잃지 말고 gate를 열어야 함	Cheonsul domain/machine slice IMPLEMENTED; release-near Human NOT_RUN
5-15분	reward/shop/route/Fate로 다음 학교를 준비	unvisited route preview, pending/commit reasons	preview는 power가 아니며 commit이 중요	domain partial; direct Workbench play BLOCKING_GAP
15-20분	네 유파를 다른 순서로 풀며 build를 누적	4th Boss → final_binding_eligible	선택 순서와 공간 build의 귀결	DOCUMENTED; current runtime not proven
true Run end	rank + distinct Boss Soul + legend conclusion	clear breakdown, persistent wallet settlement	다음 Run에서 쓸 한 번의 retry/수평 해금	DOCUMENTED; final package deferred

04. Loops and Game Flow

Core Loop — SYS-COMBAT-001

```
flowchart LR
  A[Read threat, distance, group] --> B[Move / position]
  B --> C[Automatic school action]
  C --> D[Visible reaction, barrier, mark, or close-range payoff]
  D --> E[Survive / kill / collect ordinary feedback]
  E --> A
```

School Session Loop — SYS-ENCOUNTER-002

```
flowchart LR
  A[Select unvisited school] --> B[Core pressure]
  B --> C[~3m Elite]
  C --> D[Chest token + Trace AVAILABLE]
  D --> E[Recover Trace]
```

```
E --> F[Boss warning / dual gate]
F --> G[~5m school Boss]
G --> H[Result + Boss reward]
H --> I[Persistent Workbench]
I --> J[Provisional next route + Fate + atomic commit]
J --> A
```

Meta Loop — SYS-META-003

```
true Run end
-> distinct school Boss eligibility + progress rank
-> one Ninja Soul settlement
-> horizontal unlock / hint / option (not permanent raw-stat dominance)
-> next Run, with optional one-Soul checkpoint retry if earned
```

Four-school Run Flow — UX-FLOW-001

```
start school
-> choose one unvisited school
-> Core / Elite / Trace / Boss
-> Boss reward -> Workbench -> Fate + provisional next route -> atomic commit
-> repeat until all four exactly once
-> Final Binding Workbench
-> separate final calamity Boss
-> true final Result / Ninja Soul / legend
```

~20 minutes means active combat through the fourth school Boss; it excludes Final Binding preparation and final calamity time.

05. System Registry

ID	system	owner / principal paths	status	next proof
SYS-COMBAT-001	movement, automatic combat, DDD	PlayerController, AutoAttackController, CombatDDDTracker, CombatResolver	IMPLEMENTED / automated	human readability
SYS-SCHOOL-002	4 school runtime host	SchoolRuntimeHost, *_runtime.gd	shallow runtime IMPLEMENTED	each school lifecycle
SYS-ENCOUNTER-003	Core→Elite→Trace→Boss	StageEncounterState, CheonsulVerticalSliceController, catalog	domain IMPLEMENTED; Cheonsul runtime partial	four-school runtime
SYS-ROUTE-004	four-school route and clear order	RunRouteState	IMPLEMENTED / automated	actual circuit
SYS-WORKBENCH-005	backpack, reward, route/Fate commit	BackpackState, RestBackpackSession, RestCommitCoordinator, RestFlowUI	domain IMPLEMENTED; UI gap	mouse/key/touch complete path

ID	system	owner / principal paths	status	next proof
SYS-ECONOMY-006	Run GOLD / Shop / chest	RunBuildState, ShopController, RestRewardController	legacy baseline IMPLEMENTED; DEC-033 mismatch	revised reward runtime
SYS-SOUL-007	persistent Ninja Soul / retry / settlement	future NinjaSoulWallet, RunSettlementLedger	DOCUMENTED	save + Result verification
SYS-RESULT-008	contribution result / next action	CombatContributionTracker, RestFlowUI	IMPLEMENTED for segment result	final settlement result
SYS-VISUAL-009	screen/asset grammar	visual canon + actual Sprite2D consumers	PARTIAL	target-resolution live review
SYS-ACCESS-010	help, focus, pointer/key/gamepad/touch	SchoolSelectionUI, HUD, Rest UI	machine contracts partial	Human/device test

06. System Specifications

SYS-COMBAT-001 — 이동과 자동전투

왜 존재하는가. 플레이어가 조준 부담 없이 적 거리·군집·위험 지역을 읽고 위치로 자신의 해법을 만들게 한다. 목표 감정은 “손은 단순하지만 나는 전장을 조종하고 있다”다.

어떻게 플레이하는가. Player는 input으로 이동한다. 선택 유평 runtime이 자동 행동을 내고, Combat Resolver가 committed modifier를 반영한다. `ui_accept`은 궁극기 요청으로 쓰이며, `combat/help/modal state`가 입력을 막을 수 있다.

필요 콘텐츠. player silhouette, generic enemy roles, projectile/field/reward feedback, compact status icon, struck-target-only HP reveal, ultimate readiness feedback. 현재 text badge/HP display는 아직 canonical visual rule과 불일치한다.

Godot 구조. `scenes/main/main_scene.tscn`의 `Player`, `WaveSpawner`, `SchoolRuntimeHost`, `CombatDDD`, `GameState`, `HUD`가 `MainController`에 연결된다. `RunBuildState`의 committed modifier snapshot만 `item/spatial combat authority`다.

완료 판단. normal/enemy/boss damage, school action, no double modifier, input guards, reaction feedback은 automated test로; 실제 resolution/HUD overlap/input feel은 live/Human validation으로 별도 확인한다.

SYS-SCHOOL-002 — 네 유평의 위험 처리

CNT ID	school	player action / meaningful choice	visible payoff	protected distinction	status
CNT-SCH-01	봉마류	안전한 이동 동선에 식신/결계를 유지	이동형 전선, support fire	stationary tower defense 금지	shallow runtime IMPLEMENTED

CNT ID	school	player action / meaningful choice	visible payoff	protected distinction	status
CNT-SCH-02	천술류	군집과 위치로 준비 조건을 만든다	blue WET seal → amber/orange SHOCK chain	direct spell aim 없이 spatial agency	runtime WET/SHOCK exists; DEC-027/028 effect DOCUMENTED
CNT-SCH-03	귀인류	근접에 오래 남을지 이탈할지 판단	close-range sustain/power payoff	low HP one-rule 금지	shallow runtime IMPLEMENTED
CNT-SCH-04	흑영류	위험 표적이 먼저 죽게 위치/빌드를 고른다	mark/execution priority result	manual hard target picking 금지	shallow runtime IMPLEMENTED

Each school needs Core x3 / Elite x1 / Boss x1 composition over shared primitives. The existing `EncounterCatalog` and profiles document the data language; actual generic scene wiring does not yet prove those compositions.

SYS-ENCOUNTER-003 — Elite, Trace, Boss

Entry/transition. `StageEncounterState` owns lifecycle state, monotonic elapsed time, one-time signals, and spawn permission; `WaveSpawner` is actuator, not gate owner. Current authored timing evidence uses Elite around 165–180 seconds and Boss gate around 260–270 seconds.

Rules. Elite kill grants chest token and non-expiring Trace AVAILABLE; Trace has no ORB/STYLE/GOLD/direct power effect; Boss requires Elite + recovered Trace + time + warning. Boss clear stabilizes the school and opens its acquisition package; power still needs reward acquisition → spatial placement → resolution → commit.

Known implementation gap. Current Cheonsul runtime accepts R/button recovery, while canonical behavior requires an actual recovery presentation. The next contract must preserve state ownership and replace only consumer/input behavior.

SYS-WORKBENCH-005 — 공간 백팩, 보상, Fate, route

Player promise. “가져온 것을 어디에 놓고 무엇을 감수할지”가 다음 전투의 인법이 된다.

Rules. Fixed 6×6 board, centered 4×3 starting active cells; purchased bags enlarge usable cells; 90° item/bag rotation; orthogonal adjacency; selected L/T bag shapes; six-slot REST buffer; preview gives zero power; regular and whole-layout mode have explicit legality. Boss/Shop/Chest acquisition must be atomic and Boss reward remains mandatory pending until selected.

Atomic boundary. Final backpack snapshot + one Fate + provisional unvisited route commit together through `RestCommitCoordinator`. UI renders snapshots and emits intent; it never owns geometry, economy, route, Fate, or combat logic.

Actual status. T01–T13 domain/presentation contracts are merged, but current Workbench has no complete Boss reward selection, board placement, rotation, combination, or touch/key/mouse path. Treat this as BLOCKING_GAP, not a cosmetic task.

SYS-ECONOMY-006 and SYS-SOUL-007 — Run GOLD and true-end Ninja Soul

resource	lifecycle	earns	spends	current contract
GOLD	Run-only	normal: recommended 20% × 1G; Elite: 5G; Boss: 10G	Shop / chest / backpack decisions	CONFIRMED, runtime mismatch
Ninja Soul	persistent	true Run-end only: each distinct school Boss 2 + rank C/B/A/S 0/1/2/4	once-per-Run same-school retry: 1	DOCUMENTED, no wallet yet

Rank is boss progress, not raw score: C=0 Boss, B=1, A=2–3, S=4. No Soul is paid after the first validation endpoint of fourth-Boss `final_binding_eligible`; that is not the true conclusion.

`RunSettlementLedger` must preserve eligible distinct Boss IDs across a retry without restoring failed-school transient items/Trace/GOLD, and must write settlement exactly once on a true ending. First wallet balance is 0; there is no free retry.

SYS-RESULT-008 — feedback and failure learning

Current `RestFlowUI/ResultView` displays contribution damage/healing/defense/status/kills/combo/GOLD/hints. Future true-end Result adds rank, distinct Boss count, Boss Soul, rank bonus, total, and a clear non-duplicate settlement outcome. A loss must identify its immediate threat and offer either valid one-Soul checkpoint retry or ordinary new Run; it must never disguise retry eligibility.

07. Content Registry

ID	content	purpose	required state	implementation truth
CNT-EN-01	generic Core enemy pool	baseline pressure/readability	Core behavior/visuals	generic <code>EnemyBasic</code> IMPLEMENTED
CNT-EN-02	school Core x3 × 4	school-owned pressure combinations	documented patterns/profiles	data IMPLEMENTED, runtime NOT_PROVEN
CNT-EL-01	school Elite x1 × 4	test risk philosophy at ~3m	telegraph + chest/Trace	Cheonsul generic representation partial
CNT-BS-01	school Boss x1 × 4	~5m school capstone	gate/result/reward	Cheonsul generic StageBoss partial
CNT-IT-01	19 base acquisition IDs	build vocabulary	lane eligibility and modifiers	catalog/domain IMPLEMENTED
CNT-CB-01	3 first-tier combinations	reward spatial thinking	atomic combine result	domain IMPLEMENTED
CNT-BG-01	5 purchasable bags	board expansion trade-off	shape/rotation/legality	catalog/domain IMPLEMENTED
CNT-FT-01	Fate candidates	benefit/cost route commitment	one pending Fate	controller/domain IMPLEMENTED
CNT-FINAL-01	Final Binding + calamity	full-run payoff	4-school support and ending	DEFERRED_BY_DECISION

08. UI/UX and Input Contract

UI ID	screen	consumer	player question	current state
UI-01	School Selection	SchoolSelectionUI	"Which danger style do I begin with?"	four choices + help IMPLEMENTED
UI-02	Combat HUD	HUD	"What threatens me; what can I do now?"	IMPLEMENTED , visual grammar gap
UI-03	Current School Help	HelpDialog	"What do I watch and try?"	optional path IMPLEMENTED ; copy expansion deferred
UI-04	Result	RestFlowUI/ResultView	"What happened and what is next?"	segment result IMPLEMENTED ; final Soul result deferred
UI-05	Shop / Fate	RestFlowUI	"What do I buy or sacrifice?"	basic controls/domain IMPLEMENTED
UI-06	Workbench / route	RestFlowUI/WorkbenchView	"What build and next risk do I commit?"	route/Fate view partial; placement gap
UI-07	Game Over / Retry	HUD/GameOverPanel	"What did I learn; can I retry?"	full Run scene reload only

Interaction requirements

- Mouse/pointer, keyboard/gamepad focus, and touch must each complete the core selection, help, combat, Workbench, retry, and Result path.
- Help is optional. It explains actual risk handling → visible information → positioning attempt → success/next goal; it cannot turn unassisted 30-second comprehension into a pass.
- Combat statuses use compact, silhouette-distinct icons. Text-only persistent badges are invalid.
- Enemy HP bars are hidden by default and shown only for the recently hit target. Exact duration requires Human observation.
- Player-facing slice hides/isolates TEST Elite/Boss shortcuts and English debug language.

09. Visual Bible and Asset Consumers

Global grammar — AST - GRAMMAR - 01

- Presentation: dark moonlit, premium painterly anime, ink/brush/calligraphy framing.
- Gameplay: animation-forward 2–3-head SD anime with restrained painterly/ink DNA.
- Base palette: black, deep navy, charcoal, restrained red, warm gold.

- School accents: Bongma gold, Cheonsul blue + amber/orange, Guiin red, Heukyeong purple/black.
- One fixed ninja face/hair/body/core outfit; school traces add layers. Only starting/main school uses strongest Trace Stage 3.

Runtime asset matrix

AST ID	asset	exact consumer	state
AST-01	player Move/Attack/Hit	Player/Visual	approved source; runtime state coverage PARTIAL
AST-02	cursed lantern/shadow beast/flame ninja	EnemyBasic/Visual variant pool	IMPLEMENTED consumer
AST-03	Cheonsul stage boss	StageBoss/Visual	IMPLEMENTED consumer
AST-04	talisman projectile	ProjectileBasic/Visual	IMPLEMENTED consumer
AST-05	golden RewardOrb	RewardOrb/Visual	IMPLEMENTED consumer
AST-06	Bongma familiar	BongmaFamiliar/Visual	consumer exists; other-school slice deferred
AST-07	moonlit battlefield	Main/BattlefieldBackdrop	IMPLEMENTED consumer
AST-08	Cheonsul flame field	Cheonsul/FlameFieldVisual	IMPLEMENTED consumer

Existing assets are durable only when repository source + SHA-256/provenance manifest + explicit LOCK state + actual consumer are all present, followed by applicable import/runtime evidence. Existing project assets are not regenerated merely for a new sheet.

DEC-034 image workflow

No image is generated merely at chat start or from an unscoped visual gap. Once a concrete runtime consumer, screen-reference, or planning board has been identified and fresh-read, one candidate may be generated without a pre-generation approval question. User then chooses LOCK, REVISE, or REJECT; only LOCK may promote it to project asset/direction lock. Candidate image text never owns gameplay truth.

10. Audio Contract — AUD-001

No approved runtime audio binaries or AudioStream consumer map currently exists. Audio is DOCUMENTED, not IMPLEMENTED.

Required future feedback families: select/confirm/cancel focus, school action, telegraph/warning, Trace recovery, Boss defeat, reward acquisition, Workbench legality/commit, Fate choice, Game Over/retry, and true-end settlement. Audio must reinforce—not be the only carrier of—state information; mute/reduced motion must not change rules.

11. Godot Architecture

Main Scene map

```

Main (Node2D, MainController)
■ ■ GameState
■ ■ CombatDDD
■ ■ BattlefieldBackdrop
■ ■ Player
■ ■ ■ Visual
■ ■ ■ Camera2D
■ ■ ■ AutoAttack
■ ■ WaveSpawner
■ ■ SchoolRuntimeHost
■ ■ ■ Bongma
■ ■ ■ Cheonsul
■ ■ ■ Guin
■ ■ ■ Heukyeong
■ ■ generic EnemyBasic instances
■ ■ HUD
■ ■ SchoolSelectionUI

```

MainController dynamically ensures RunBuildState, ShopController, FateController, StageFlowController, CombatContributionTracker, CombatResolver, and RestFlowUI. This is existing architecture, not a recommendation to make UI own domain state.

Authority and event flow

```

flowchart LR
    CAT[definitions/catalog] --> BP[BackpackState]
    BP --> RES[BackpackResolver]
    RES --> REST[RestBackpackSession / CombinationResolver]
    REST --> COMMIT[RestCommitCoordinator]
    COMMIT --> BUILD[committed RunBuildState]
    BUILD --> COMBAT[CombatResolver / Player / SchoolRuntimeHost]
    ROUTE[RunRouteState] --> COMMIT
    ENCOUNTER[StageEncounterState] --> MAIN[MainController actuator]
    MAIN --> UI[HUD / RestFlowUI]
    UI -->|intent only| COMMIT

```

Principal data contracts

DAT ID	owner	key facts	protected rule
DAT-01	BackpackState	cells, instances, rotation, active area	no UI/economy/combat authority
DAT-02	BackpackResolver	adjacency, previews, modifier resolution	deterministic and read-only
DAT-03	RestBackpackSession	buffer, edits, undo/redo, whole-layout mode	preview zero combat power

DAT ID	owner	key facts	protected rule
DAT-04	RunBuildState	GOLD, selected school, committed modifier snapshot	singular item/spatial combat authority
DAT-05	RunRouteState	unvisited, active/provisional, clear order, stage	school identity and stage axes separate
DAT-06	StageEncounterState	Core/Elite/Trace/Boss lifecycle	no economy/route/UI ownership
DAT-07	future RunSettlementLedger	distinct Boss eligibility, one-time settlement	survives retry without duplicate payout
DAT-08	future NinjaSoulWallet	persistent balance	only persistent owner writes balance

Save/load and platform

Current slice has no persistent save/profile/Wallet owner. Ninja Soul introduces a bounded persistent save requirement but does not authorize a broad meta-power system. Target engine render method is Godot 4.7 [gl_compatibility](#) for desktop/mobile configuration. Release title/save/settings/loading, Android export, performance target, pause, reduced-motion, and device QA remain production requirements, not completed features.

12. Test and QA Contract

QA ID	validation	current evidence	cannot prove
QA-01	unit/domain GUT	T01-T16 and current test files	live readability/fun
QA-02	integration/main scene	existing Main, stage, HUD, Rest tests	end-to-end new four-school run
QA-03	import/parse/headless smoke	prior exact PR evidence	Human/player/device
QA-04	live render/input	narrow historical runtime packages only	current full slice visual UX
QA-05	Human usability	NOT_RUN	do not infer from automation
QA-06	Player experience	NOT_RUN	do not infer from assets or docs
QA-07	touch/gamepad/device/export	NOT_RUN	focus contract alone is insufficient

First Human vertical slice acceptance

1. Players can describe the selected school's danger-handling verb without opening help; optional help is evaluated separately.
2. Core→Elite→Trace→Boss timing, telegraph fairness, Trace purpose, and Boss gate are readable.
3. Workbench spatial placement, rotation, reward selection, Fate, route preview, and atomic commit are completable through pointer, keyboard/gamepad, and touch.

4. Status icons, one-hit HP reveal, Korean copy, target resolution, effect density, and Boss warning remain legible.
5. The run can traverse all four schools exactly once and fourth Boss makes `final_binding_eligible` legible. It does not claim final ending or Soul settlement.

13. Benchmark and Positioning — researched 2026-08-28

reference	observed fit	ADOPT / ADAPT / REJECT	project-specific conversion
Vampire Survivors	time survival, escalating hordes, gold between runs	ADAPT	fast threat escalation and failure→next-attempt meaning; do not copy weapon/level-up surface or raw stat meta
Brotato	short waves, item shop, accessibility controls	ADAPT	clear inter-wave decision cadence and readable trade-offs; preserve continuous school encounter rather than copying waves/items
Soulstone Survivors	horde action, bosses, build synergies	REFERENCE_ONLY	Boss capstone and synergy readability are useful benchmarks; reject content-volume race and 3D spectacle expectation
Hades II	authored rogue-like encounters and memorable dark-fantasy identity	ADAPT	consequence-rich route/failure tone and coherent fantasy presentation; reject manual combat scope and narrative production burden
Backpack Battles	arrange items, buy/craft, watch auto battle	ADAPT	spatial placement must visibly change build and be explainable; reject PvP/rank structure and copyable item surface

Positioning statement. ■■■ ■ sits between top-down auto-action survival and deliberate inventory/spatial buildcraft. Its hook is not “another bullet heaven” or “another backpack auto-battler”; it is the transformation of four ninja danger philosophies into one fixed protagonist’s readable field behavior and atomic spatial build.

14. Roadmap, Dependencies, and Risks

order	package	player value / dependency	primary risk
Q-01	Phase 1 completion + unified implementation contract	prevent canon/code/UI drift before build	hidden current decisions
Q-02	Trace/telegraph + combat information grammar	fair, readable encounter learning	target-resolution failure
Q-03	complete spatial Workbench inputs	turn planning promise into play	UI/domain authority leakage
Q-04	four-school shared lifecycle/content composition	validate selection/route promise	four separate engines/content cost
Q-05	DEC-033 Run economy, wallet, retry, true-end settlement	meaningful failure and horizontal future	duplicate save/settlement payout

order	package	player value / dependency	primary risk
Q-06	Human vertical slice	prove player promise	machine evidence mistaken for UX
Q-07	Final Binding/final calamity	deliver full legend payoff	premature scope expansion

Evidence-based SWOT

class	statement	evidence / confidence	player impact	disposition
STRENGTH	spatial backpack already has protected deterministic domain boundaries	T01-T12 code/tests; VERIFIED for domain	choices can be legal, explainable, and committed once	PROTECT
STRENGTH	four schools have distinct approved verbs	canon/DEC-027/028; CONFIRMED	supports memorable starts and routes	PROTECT
WEAKNESS	actual playable lifecycle is Cheonsul-only and generic in visuals/content	MainController , scenes; VERIFIED	current selection overpromises	IMPROVE
WEAKNESS	Workbench direct input is incomplete	RestFlowUI ; VERIFIED	core build promise cannot be judged	IMPROVE
WEAKNESS	Human/player/device evidence is absent	evidence ledger; NOT_RUN	readability/fun unknown	TEST
OPPORTUNITY	survival + spatial auto-build market language is legible	official references; INFERENCE	concept can be marketed in one sentence	TEST
THREAT	content/visual complexity can turn four schools into four games	DEC-026 budget + actual gaps; VERIFIED	schedule and identity risk	MITIGATE
THREAT	raw score/extra effects can obscure school identity	canon visual/information findings; PARTIAL	player cannot learn intended choice	MONITOR

15. User Decision Required

No new product-meaning decision is required to write or use this GDD. Values explicitly marked **TUNE_RECOMMENDED** (normal 20% gold chance, Soul reward table) require Human/Player balance evidence before adjustment. The next decision gate is the unified implementation contract's Definition of Ready review, not a new feature invention.

16. Change Log

date	change	authority
2026-08-28	initial Master GDD created from current main	this document
2026-08-28	true Run-end-only Ninja Soul settlement confirmed	DEC-033 + user approval

date	change	authority
2026-08-28	generate-one-candidate then user final lock workflow confirmed	DEC-034 + user approval
2026-08-28	historical Notion Home retry wording and Visual Bible generation cadence were reconciled before migration	historical receipts + DEC-033/034
2026-08-28	former Notion structure/current work products preserved as a repository migration archive before repository-only cutover	DEC-035 + migration manifest
2026-08-28	사람용 다운로드 PDF와 재생성/검수 계약을 추가	user-approved repository PDF publication

17. Non-negotiable Evidence Ceiling

This document is **DOCUMENTED** and parts of its linked domain are **IMPLEMENTED/AUTOMATED_TEST_PASS**. It does **not** prove a fun four-school Run, a finished Workbench, final settlement, visual target-resolution readability, Human usability, Player Experience, touch/gamepad, Android/export, or release readiness.